CHAPTER 1

INTRODUCTION

Cryptocurrency is a digital or virtual form of currency that uses cryptography for security, making it resistant to counterfeiting. Unlike traditional currencies issued by governments, cryptocurrencies operate on decentralized networks based on blockchain technology. Bitcoin, created in 2009 by an unknown person or group using the pseudonym Satoshi Nakamoto, was the first and remain the most well-known cryptocurrency. Other popular cryptocurrencies include Ethereum, ripple, and Litecoin. The decentralized nature of cryptocurrencies allows for peer-to-peer transactions without the need for intermediaries like banks. Their vale can be volatile, influenced by factors such as market demand, technological developments, and regulatory changes. Decentralization means cryptocurrencies operate on decentralized networks, reducing dependence on central authorities like government or banks. Security: Cryptography ensures secure transactions, protecting against fraud and providing users with control over their funds. Accessibility means Cryptocurrencies enable financial inclusion by allowing anyone with internet access to participate, regardless of geographic location or traditional banking infrastructure. Efficiency means Transactions can be processed quickly, especially in comparison to traditional banking systems, which may involve delays. Lower Transaction Costs means Cryptocurrency transactions often have lower fees compared to traditional financial services. Privacy means Cryptocurrencies offer varying degrees of privacy, allowing users to conduct transactions with a certain level of anonymity. Volatility means Cryptocurrency values can be highly volatile, leading to substantial price fluctuations within short periods. Regulatory Uncertainty means Regulatory frameworks for cryptocurrencies are still evolving, leading to uncertainty and potential legal challenges. Security Concerns means While the underlying blockchain technology is secure, the infrastructure surrounding cryptocurrencies, such as exchanges, can be vulnerable to hacks and cyber attacks. Limited Acceptance means Despite growing acceptance, many businesses and individuals still do not accept cryptocurrencies as a valid form of payment. Irreversibility of Transactions means Once confirmed, cryptocurrency transactions are typically irreversible, which can be problematic in the case of accidental or fraudulent transactions. Environmental Impact means Proof-of-work cryptocurrencies, like Bitcoin, require significant computational power, leading to concerns about their environmental impact due to high energy consumption.Understanding these pros and cons is crucial for individuals considering involvement with cryptocurrencies

CHAPTER 2

LITRATURE SURVEY

In one of works related to Cryptoguard, is by Fengyin Li, Xinying Yu, Rui Ge, Yanli Wang, Yang Cui and Huiyu Zhou. In their paper, they have addressed the issues with the identity authentication using Public Key Infrastructure (PKI) and also issues with not having any security for password guessing attack. In this paper, the security issues and identity authentication issues are sorted. PKI is a framework that manages digital keys and certificates. It involves the use of public and private key pairs to secure communication and verify the identity of users. The authors may have addressed security issues related to identity authentication using PKI. Common challenges include certificate management, key distribution, and ensuring the integrity of the PKI infrastructure. The authors likely propose solutions or mechanisms to address the security issues and identity authentication challenges discussed in the paper.

The other work related to Cryptoguard is by Katja Tuma, Romy Van Der Lee. In this paper, they have addressed issues like biased judgement. The model in this paper improves the security of the existing system, better risk analysis, threat impact detection, integrity, accountability and availability. However, this paper is an ideal paper and there is no model made. It can be implemented but, all the advantages can not be implemented in one model at once. The paper addresses issues related to biased judgment, which could refer to situations where subjective or prejudiced assessments impact security-related decisions. Biased judgment can introduce vulnerabilities and weaken overall security. The proposed model likely includes mechanisms for detecting and assessing the impact of security threats. This is crucial for understanding the severity of potential incidents.

In the idea given by Yunyeong Goh, Jusik Yun, Dongjun Jung and Jong-Moon Chung, they have addressed issues are blockchain scalability, security issues, safety issues and data management issues. This model improves scalability, latency, guarantees safety and maximises transactions per second. Scalability is a crucial challenge in blockchain systems, especially as the number of transactions and participants grows. It refers to the system's ability to handle an increasing workload efficiently. The authors likely propose solutions to enhance scalability, such as sharding, layer 2 solutions, or other techniques that allow the blockchain network to process a higher number of transactions. Safety in the context of blockchain may involve ensuring the integrity and reliability of the data and transactions on the network. It could also include measures to prevent double-spending or fraudulent activities. Latency refers to the time it takes for a transaction to be confirmed. The model likely includes mechanisms to reduce transaction confirmation times.

The other idea by Henri T. Heinonen, Alexander Semenov, Jari Veijalainen and Timo Hamalainen. The paper addresses the issues related to high energy consumption during transactions. They have found the solution to this problem in the paper but the solution can lessen the consumption of energy only to an extent. Security is paramount in blockchain systems, as they often deal with sensitive and valuable data. Security issues could include vulnerabilities in smart contracts, the risk of 51% attacks, or weaknesses in consensus algorithms. The proposed model likely introduces improvements or novel approaches to enhance the overall security of the blockchain network. Safety in the context of blockchain may involve ensuring the integrity and reliability of the data and transactions on the network. It could also include measures to prevent double-spending or fraudulent activities. The model is likely designed to provide safety guarantees, ensuring that transactions are secure, consistent, and tamper-resistant. Blockchain networks often face challenges related to the efficient management of data, including storage, retrieval, and querying. As the size of the blockchain grows, data management becomes a critical consideration. Latency refers to the time it takes for a transaction to be confirmed. The model likely includes mechanisms to reduce transaction confirmation times.

2.1 PROBLEM STATEMENT

De-anonymization methods are employed to monitor and track illegal activities performed using cryptocurrency transactions.

Cryptocurrency transactions are recorded on a public blockchain, providing transparency. Analysts and investigators can examine transaction histories using tools like blockchain explorers to trace the flow of funds.

Analysts link multiple addresses to a single user, creating clusters of addresses associated with the same entity. This helps reveal patterns of behavior and provides insights into the user's activities.

Identifying unique transaction patterns or behaviors, such as timing, amounts, and frequency, assists in linking transactions to specific users. This type of analysis helps in understanding the modus operandi of individuals.

Since cryptocurrency transactions are recorded on a public blockchain, analysts can trace funds by examining transaction histories. Tools like blockchain explorers provide transparency, enabling investigators to follow the flow of funds.

By linking multiple addresses to a single user, analysts can form clusters of addresses associated with the same entity. This can reveal patterns of behaviour and provide insights into the user's activities.

Identifying unique transaction patterns or behaviours can help in linking transactions to specific users. This includes analysing the timing, amounts, and frequency of transactions.

Collaboration with cryptocurrency exchanges and wallet providers allows authorities to access user information, helping to link identities to specific transactions.

Connecting cryptocurrency transactions to IP addresses can provide additional information about the users involved. This method is more effective when combined with other de-anonymization techniques.

Tracking transactions through mixing services, which are sometimes used to obscure the origin of funds, can be challenging but is not impossible. Sophisticated analysis may reveal patterns that lead back to the original source.

It's important to note that privacy-focused cryptocurrencies aim to enhance user anonymity, making de-anonymization more challenging. However, law enforcement agencies continually develop methods to stay ahead of those who exploit cryptocurrencies for illicit purposes. Striking a balance between privacy and security remains a complex challenge in the evolving landscape of cryptocurrency technologies

2.2 EXISTING SYSTEM

One of the trending and most used cryptocurrencies is Bitcoin. It is also the first decentralized cryptocurrency. Nodes in peer-to-peer bitcoin network verify transactions through cryptography and record them in a public distributed ledger called blockchain, without a central oversight. Consensus between nodes is achieved using a computationally intensive system based on proof-of-work called mining.

Bitcoin mining requires increasing quantities of electricity and was responsible for 0.2% of greenhouse gas emissions as of 2022. Use of bitcoin as a currency began in 2009, with the release of its open-source implementation.

Bitcoin is currently used more as a store of value and less as a medium of exchange or unit of account. It is mostly seen as an investment and has been described by many scholars as an economic bubble. As bitcoin is pseudonymous, its use by criminals has attracted the attention of regulators, leading to its ban by several countries as of 2021.

As a decentralized system, bitcoin operates without a central authority or single administrator so that anyone can create a new bitcoin address and transact without needing any approval. This is accomplished through a specialized distributed ledger called a blockchain that records bitcoin transactions.

In the blockchain, bitcoins are linked to specific addresses that are hashes of a public key. Creating an address involves generating a random private key and then computing the corresponding address. This is almost instant, but the reverse is nearly impossible. Losing a private key means losing access to the bitcoin, with no other proof of ownership accepted by the protocol.

The mining process in Bitcoin involves maintaining the blockchain through computer processing power. Bitcoin wallets were the first cryptocurrency wallets, enabling users to store the information necessary to transact bitcoins.

2.3 DRAWBACKS OF EXISTING SYSTEM

i. Transactions are used for illegal purposes also.
ii. More energy consumption
iii. Irreversibility of transaction.
iv. Technological complexity
v. Lack of consumer protection
vi. Due to anonymity of users, the transaction can not be tracked. Henceforth it is used for illegal purposes.

2.4 PROPOSED SYSTEM

Cryptoguard is a software which can be used to detect illegal activities that are being performed by using cryptocurrency. Cryptocurrency is a financial tool that can be misused for illegal activities due to anonymity that it provides. Not only anonymity, but also ease in cross-border transaction and its decentralized nature give an ease of access to the criminals to perform multiple illegal activities like Fraud, Tax evasion, Money laundering, Ransomware attacks, Fraudulent investment schemes, etc. The website will be able to detect multiple illegal activities like Fraud, Tax evasion, Money laundering, Ransomware attacks, Fraudulent investment schemes, etc.

ADVANTAGES OF PROPOSED SYSTEM

i. Track anonymous illegal transactions
ii. It will have less consumption of energy
iii. Users will be anonymous to each other but, their actions will be tracked. Henceforth, it will not be used for illegal purposes.

iv. Secure.
v. Even though, this product will break a part of the users anonymity, it will keep the users information private to other users.
vi. It is less complex.

2.5 OBJECETIVES

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CHAPTER 3

SYSTEM REQUIREMENT SPECIFICATION

This document details the requirements for Management System for Distributed Environment. The software requirements shall be specified for all the phases of Management System.
A Software Requirements Specification (SRS) is a complete description of the behaviour of the system to be developed.

It includes a set of use cases that describe all of the interactions that the users will have with the software. Use cases are also known as functional requirements. In addition to use cases, the SRS also contains non-functional (or supplementary) requirements. Non-functional requirements are requirements which impose constraints on the design or implementation (Such as performance engineering requirements, quality standards, or design constraints).

1.1 Functional Requirements:

The functional requirements for a system describe what the system should do. These requirements depend on the type of software being developed; the general approach taken by the organization when writing requirements. The functional system requirements describe the system function in detail, its inputs and outputs, exceptions and so on. Functional requirements are as follows:

- User Registration and Authentication
- Wallet Management.
- Transaction Processing
- Blockchain Integration
- Cryptocurrency Exchange
- Security Measures

1.2 Non-Functional Requirements:

Non-functional requirements, as the name suggests, are requirements that are not directly concerned with the specific functions delivered by the system. They may relate to emergent system properties such as reliability, response time and store occupancy. Alternatively, they may define constraints on the system such as capabilities of I/O devices and the data representations used in system interfaces. The non-functional requirements are as follows:

- Performance.
- Scalability.
- Reliability.

1.3 Hardware Requirements:

The Most common requirements defined by any operating system or software application is the physical computer resources, also known as hardware. A hardware requirement list is often accompanied by a hardware compatibility list (HCL), especially in case of operating system. An HCL lists tested, compatibility and sometimes incompatible hardware devices for a particular operating system or application. The following sub-sections discuss the various aspects of hardware requirements.

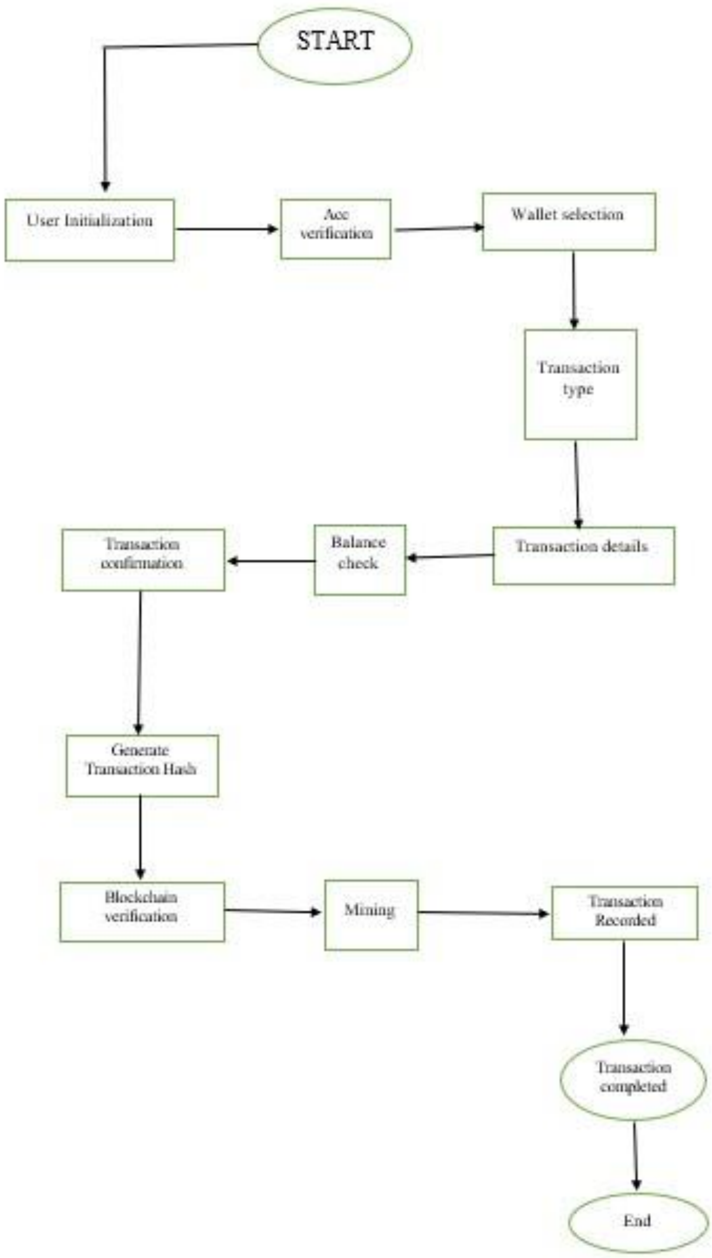
Processor	:	Intel Core
Processor Speed	:	2.5 GHz to 2.8 GHz
RAM	:	8 GB
RAM Hard Disk	:	1 TB

1.4 Software Requirements:

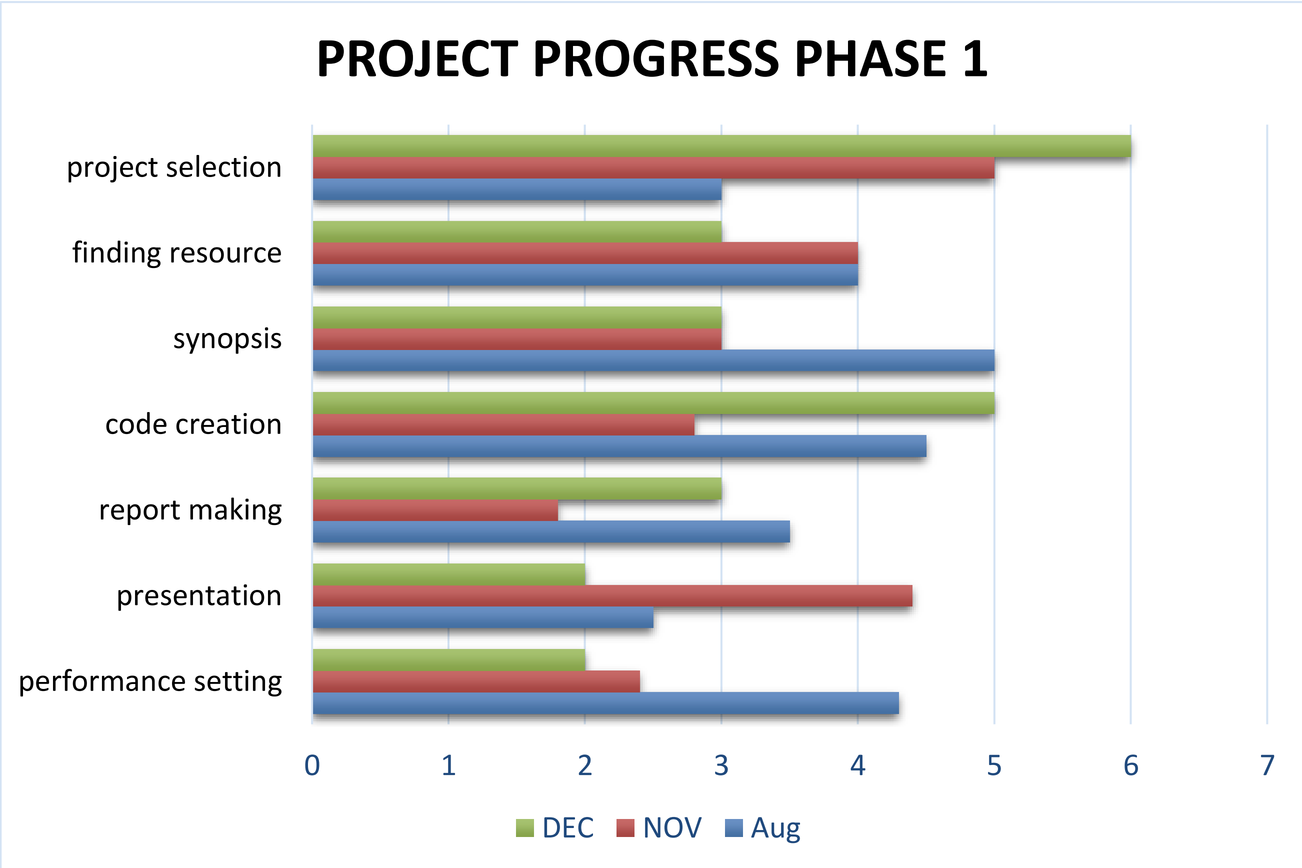
Software requirements deal with defining software resources requirements and pre requirements that need to be installed on a computer to provide optimal functioning of an application. These requirements or pre- requirements are generally not included in the software installed package and need to be installed separately before the software installed package and need to be installed separately before the software is installed.

Operating System	:	Windows 11
Database	:	MySQL
Front End	:	Html, CSS, Javascript
Back End	:	Python

FLOW CHART


START
User Initialization
Acc verification
Wallet selection
Transaction type
Transaction details
Balance check
Transaction confirmation
Generate Transaction Hash
Blockchain verification
Mining
Transaction Recorded
Transaction completed
End

GANTT CHART


PROJECT PROGRESS PHASE 1
project selection
finding resource
synopsis
code creation
report making
presentation
performance setting
0
1
2
3
4
5
6
7
DEC
NOV
Aug

REFERENCE

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